NEP Haryana Excellence Awards 2026: Server and VM Requirements

Application	NEP Haryana Excellence Awards 2026 web portal
Purpose	Request for staging and production hosting (first deployment)
Date	19-10-2026
Detailed reference	`VM_SERVER_REQUIREMENTS.pdf` (full technical assessment)

1. Application overview

- **Users:** higher-education institutions of Haryana (about 1,700 on the AISHE list), state administrators and review committees.
- **Front end:** React web application, served as static files.
- **Back end:** Python 3.12 / Django REST API, run by Gunicorn behind Nginx.
- **Database:** PostgreSQL 16.
- **Evidence files:** PDF, image, DOCX and XLSX, up to 25 MB each, stored on a dedicated disk volume.
- **Background jobs:** none.

2. Virtual machines

Sizes are a conservative starting point, to be confirmed by a load test on staging before the submission window opens.

#	Role	Env	OS	vCPU	RAM	Storage
1	Application / web server: Nginx, Gunicorn/Django, Redis	Production	Ubuntu Server 24.04 LTS, 64-bit	4	8 GB	60 GB SSD (OS + app) + 250 GB SSD separate, expandable evidence volume
2	Database server: PostgreSQL 16	Production	Ubuntu Server 24.04 LTS, 64-bit	4	8 GB	40 GB SSD (OS) + 100 GB SSD (database)
3	Staging / UAT: all components on one VM	Staging	Ubuntu Server 24.04 LTS, 64-bit	2	4 GB (8 GB preferred)	100 GB SSD
4	Backup storage, separate from production VMs	Shared	n/a	n/a	n/a	1 TB to start, expandable
	Total			**10**	**20 GB** (24 GB preferred)	**~550 GB SSD + 1 TB backup**

Alternatives: a managed PostgreSQL service can replace VM 2. If only one production VM is possible: 8 vCPU / 16 GB RAM with separate OS, database and evidence volumes.

3. Server software

Component	Version	Where
Python (with venv)	3.12	VM 1, VM 3
Nginx	OS package	VM 1, VM 3
Gunicorn (systemd service)	26.x, from the application's requirements	VM 1, VM 3
PostgreSQL server	16 (14 minimum)	VM 2, VM 3
PostgreSQL client tools	same major version	VM 1 (backups)
Redis	7.x, localhost only, password-protected	VM 1, VM 3
Certbot or state CA tooling	current	VM 1, VM 3

Node.js is **not** needed on the servers; the front end is built beforehand.

4. Network and firewall

Direction	From → To	Port	Purpose
Inbound	Internet → VM 1	443/TCP	HTTPS portal and API

Direction	From → To	Port	Purpose
Inbound	Internet → VM 1	80/TCP	Redirect to HTTPS only
Inbound	Admin VPN / bastion → all VMs	22/TCP	SSH, key-based only; not open to the internet
Internal	VM 1 → VM 2	5432/TCP	PostgreSQL
Local only	VM 1	6379, 8000	Redis, Gunicorn; never exposed
Outbound	VM 1 → mail relay	587/TCP (or 465)	Verification, invitation and password-reset emails
Outbound	All VMs → package mirrors	443/TCP	Security updates (proxy/mirror acceptable)
Outbound	All VMs → NTP	123/UDP	Time synchronisation

- Firewall, WAF or load balancer must allow request bodies of **at least 30 MB** (evidence uploads).
- Bandwidth: **at least 100 Mbps** shared uplink to start.

5. Domain, SSL and email

- One sub-domain, e.g. `[app-subdomain].haryana.gov.in`, pointing to VM 1.
- SSL/TLS certificate from a browser-trusted CA, with a renewal arrangement.
- Authenticated **SMTP relay** and an approved sender address. Sign-up cannot be completed without working email.
- **SPF, DKIM and DMARC** DNS records for the sender domain.

6. Backup, monitoring and security

- Daily PostgreSQL backups with point-in-time recovery, plus regular backups of the evidence volume, taken together and stored off the production VMs.
- Retention and RPO/RTO to be agreed.
- Monitoring of CPU, RAM and disk (alert at **70%** on the evidence volume), uptime checks and certificate-expiry alerts.
- OS hardening and regular security patching; antivirus scanning of the evidence volume if policy requires it.
- Passwords and keys are exchanged through a secure channel, never by email.

7. Access needed by the project team

- SSH with sudo on VM 1 and VM 3 via VPN or bastion, key-based.
- Database administrator access on VM 2 (or the managed database) to create the application database and user.
- An agreed deployment method: SSH from CI, or scheduled manual windows.

8. Points for confirmation

1. Hosting facility, and whether Ubuntu 24.04 LTS is permitted (or RHEL/Rocky 9).
2. Availability of managed PostgreSQL, managed backups, load balancer or WAF.
3. SMTP relay and sender address; who manages the sender domain's DNS.
4. Backup retention and off-site copy requirement.
5. Approved access and deployment method.
6. Whether a security audit (e.g. a CERT-In empanelled auditor) is required before go-live.
7. Log retention, and whether logs go to a central SIEM.

9. Requested sequence

1. **Staging VM (VM 3) first**, for installation, email and load testing.
2. Production VMs (VM 1, VM 2) and backup storage after staging is validated.